

CS4084

Quantum Computing

Week 10 – Lecture 19-20

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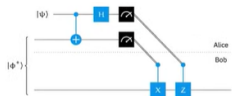
Quick Review: What We Covered Last Lecture I

- **Quantum Circuits:** Graphical model of quantum evolution
 - **Wires** $\rightarrow$ Qubits **Gates** $\rightarrow$ Unitary operations
 - **Local vs Global Operations:**
 - Local: Act on single qubit ($H \otimes I$)
 - Global: Multi-qubit (e.g., CNOT) $\Rightarrow$ enables **entanglement**
- **Inner Product & Geometry:**
 - $\langle \phi | \psi \rangle$ measures similarity ("overlap")
 - **Orthogonality:** $\langle \phi | \psi \rangle = 0$
 - **Orthonormal Basis:** $\langle v_i | v_j \rangle = \delta_{ij}$
 - $\Rightarrow$ Forms the basis for quantum measurement
- **Projection Operators:**
 - $P = |\psi\rangle\langle\psi|$
 - Used to extract probabilities from quantum states
- **Compound Systems & Tensor Products ($\otimes$):**

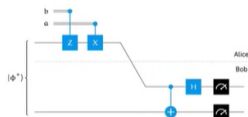
Quick Review: What We Covered Last Lecture II

- Multi-qubit space: $(\mathbb{C}^2)^{\otimes n}$
- Basis ordering: $|00\rangle, |01\rangle, |10\rangle, |11\rangle$
- Operations combine via **Kronecker product**
- **Bilinearity:**
 - $\alpha(|A\rangle \otimes |B\rangle) = (\alpha|A\rangle) \otimes |B\rangle$
 - Enables expansion of superposition states
- **Probabilistic Interpretation:**
 - Measurement outcomes derived using projections
 - Probabilities given by squared amplitudes
- **Key Connection to Today:**
 - Multi-qubit systems allow **correlations beyond classical physics**
 - $\Rightarrow$ Today: **Entanglement, Quantum Communication, and Nonlocality**

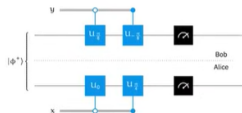
Quantum Teleportation, Superdense Coding, and CHSH Game



Teleportation



Superdense coding



CHSH game

Why Entanglement Matters

Key Insights

- Entanglement enables **powerful quantum communication protocols**
- Information can be transmitted in **non-classical ways**
- Leads to phenomena that **surpass classical limits** (e.g., CHSH game)

Entanglement is the key resource behind quantum advantage

What is Quantum Entanglement?

- Entanglement is a uniquely **quantum correlation** between qubits
- A joint state that **cannot be written as a product**:

$$|\psi\rangle \neq |a\rangle \otimes |b\rangle$$

- Example (Bell State):

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

- **Key Property:**
 - Measuring one qubit instantly determines the other
- **Important:**
 - No classical system can reproduce this behavior

Measurement in Entangled Systems

Consider the Bell state:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

- If Alice measures:

- Gets 0 $\Rightarrow$ Bob's state becomes $|0\rangle$
- Gets 1 $\Rightarrow$ Bob's state becomes $|1\rangle$

- **Observation:**

- Outcomes are perfectly correlated
- But each result is individually random

- **Key Insight:**

- Correlation exists **without classical communication**

This is the foundation of teleportation and superdense coding

Entanglement Outcomes

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Case 1

$|00\rangle$

Alice: 0 Bob: 0

Case 2

$|11\rangle$

Alice: 1 Bob: 1

- Measurement randomly selects one branch
- Correlation is preserved in both cases

Final remarks on Entanglement

In Lesson 2, we encountered this example of an *entangled state* of two qubits:

$$|\phi^+\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$

It is typical in the study of quantum information and computation that we view entanglement as a *resource* that can be used to accomplish different tasks.

When we do this we view the state $|\phi^+\rangle$ as representing one unit of entanglement called an *e-bit*.

Terminology

To say that Alice and Bob *share an e-bit* means that Alice has a qubit A, Bob has a qubit B, and together the pair (A, B) is in the state $|\phi^+\rangle$.

Quantum Teleportation: What and Why

- **What is Quantum Teleportation?**

- A process to transfer an unknown quantum state from one location (Alice) to another (Bob) without physically sending the particle.
- Relies on **entanglement** and classical communication.

- **Why is it important?**

- Enables secure quantum communication and forms the basis of a **quantum internet**.
- Demonstrates non-classical phenomena that cannot be achieved in classical physics.
- Essential for other quantum protocols like **superdense coding** and distributed quantum computation.

Teleportation Set-up

Scenario

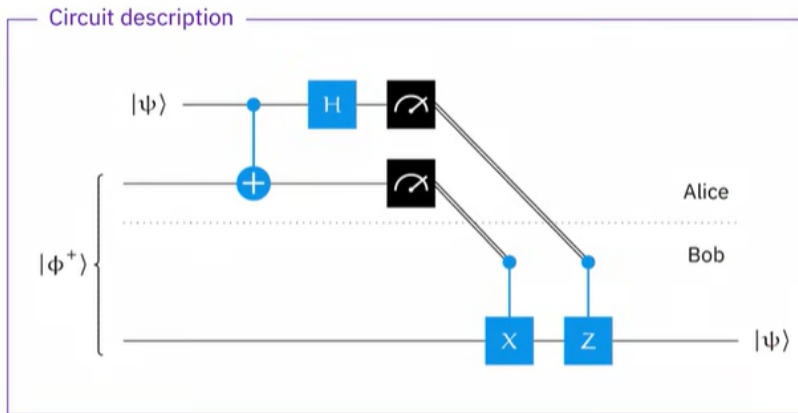
Alice has a **qubit** Q that she wishes to transmit to Bob.

- Alice is unable to physically send Q to Bob — she is only able to send **classical information**.
- Alice and Bob **share an e-bit**.

Remarks

- The state of Q is “unknown” to both Alice and Bob.
- Correlations (including entanglement) between Q and other systems must be preserved by the transmission.
- The **no-cloning theorem** implies that if Bob receives the transmission, Alice must no longer have the qubit in its original state.

Teleportation Circuit Diagram



Functional Roles of Gates in Teleportation

- **Alice's Side (The Transmitter):**

- **CNOT Gate:** The *Entangler*. It links the unknown state $|\psi\rangle$ to the shared Bell pair.
- **H (Hadamard) Gate:** The *Basis Shifter*. It rotates the qubits into the Bell basis so they can be measured without destroying the quantum information.
- **Measurement (Meter):** The *Scanner*. It collapses the quantum state into two classical bits (0 or 1) to be sent to Bob.

- **Bob's Side (The Receiver):**

- **X Gate:** The *Bit-Flip Repair*. Applied only if Alice's second bit is 1. It flips $|0\rangle \leftrightarrow |1\rangle$.
- **Z Gate:** The *Phase-Flip Repair*. Applied only if Alice's first bit is 1. It fixes the $\pm$ sign of the state.

Key Takeaway

Alice **scrambles** the state to send it; Bob uses the classical bits as a "key" to **unscramble** it using X and Z gates.

Summary: Roles of Gates in Teleportation

Gate	Who?	Job (Scientific)	Analogy
CNOT	Alice	Entangles message to bridge	The Glue
H	Alice	Shifts basis for measurement	The Scrambler
Meter	Alice	Collapses state to 2 bits	The Scanner
X	Bob	Corrects Bit-Flip error	The Wrench
Z	Bob	Corrects Phase-Flip error	The Mirror

The Process in One Sentence

Alice uses **CNOT** and **H** to "upload" the data, then uses the **Meter** to destroy the original; Bob uses **X** and **Z** to "reconstruct" the data at his end.

Initial Condition and Protocol

Initial conditions

Alice and Bob share one e-bit: Alice has a qubit A, Bob has a qubit B, and (A, B) is in the state $|\phi^+\rangle$.

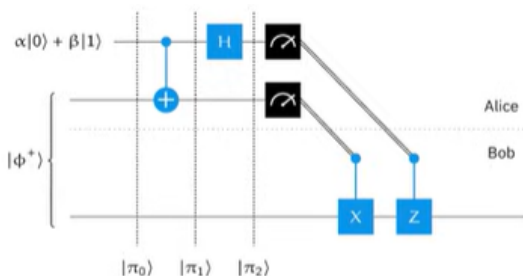
Alice also has a qubit Q that she wishes to transmit to Bob.

Protocol

1. Alice performs a controlled-NOT operation, where Q is the control and A is the target.
2. Alice performs a Hadamard operation on Q.
3. Alice measures A and Q, obtaining binary outcomes a and b , respectively.
4. Alice sends a and b to Bob.
5. Bob performs these two steps:
 - 5.1 If $a = 1$, then Bob applies an X operation to the qubit B.
 - 5.2 If $b = 1$, then Bob applies a Z operation to the qubit B.

I	if $ab = 00$
Z	if $ab = 01$
X	if $ab = 10$
ZX	if $ab = 11$

Teleportation Analysis



$$|\pi_0\rangle = \frac{\alpha|000\rangle + \alpha|110\rangle + \beta|001\rangle + \beta|111\rangle}{\sqrt{2}}$$

$$|\pi_1\rangle = \frac{\alpha|000\rangle + \alpha|110\rangle + \beta|011\rangle + \beta|101\rangle}{\sqrt{2}}$$

$$|\pi_2\rangle = \frac{\alpha|00\rangle|+\rangle + \alpha|11\rangle|+\rangle + \beta|01\rangle|-\rangle + \beta|10\rangle|-\rangle}{\sqrt{2}}$$

Expanding the last term..

$$\begin{aligned} |\pi_2\rangle &= \frac{\alpha|00\rangle|+\rangle + \alpha|11\rangle|+\rangle + \beta|01\rangle|-\rangle + \beta|10\rangle|-\rangle}{\sqrt{2}} \\ &= \frac{\alpha|00\rangle(|0\rangle + |1\rangle) + \alpha|11\rangle(|0\rangle + |1\rangle) + \beta|01\rangle(|0\rangle - |1\rangle) + \beta|10\rangle(|0\rangle - |1\rangle)}{2} \\ &= \frac{\alpha|000\rangle + \alpha|001\rangle + \alpha|110\rangle + \alpha|111\rangle + \beta|010\rangle - \beta|011\rangle + \beta|100\rangle - \beta|101\rangle}{2} \\ &= \frac{1}{2}(\alpha|0\rangle + \beta|1\rangle)|00\rangle + \frac{1}{2}(\alpha|0\rangle - \beta|1\rangle)|01\rangle + \frac{1}{2}(\alpha|1\rangle + \beta|0\rangle)|10\rangle + \frac{1}{2}(\alpha|1\rangle - \beta|0\rangle)|11\rangle \end{aligned}$$

Expending the last term..

$$|\pi_2\rangle = \frac{1}{2}(\alpha|0\rangle + \beta|1\rangle)|00\rangle + \frac{1}{2}(\alpha|0\rangle - \beta|1\rangle)|01\rangle + \frac{1}{2}(\alpha|1\rangle + \beta|0\rangle)|10\rangle + \frac{1}{2}(\alpha|1\rangle - \beta|0\rangle)|11\rangle$$

a b	Probability	Conditional state of (B, A, Q)	Operation on B	Final state of B
00	$\frac{1}{4}$	$(\alpha 0\rangle + \beta 1\rangle) 00\rangle$	$\mathbb{I}$	$\alpha 0\rangle + \beta 1\rangle$
01	$\frac{1}{4}$	$(\alpha 0\rangle - \beta 1\rangle) 01\rangle$	Z	$\alpha 0\rangle + \beta 1\rangle$
10	$\frac{1}{4}$	$(\alpha 1\rangle + \beta 0\rangle) 10\rangle$	X	$\alpha 0\rangle + \beta 1\rangle$
11	$\frac{1}{4}$	$(\alpha 1\rangle - \beta 0\rangle) 11\rangle$	ZX	$\alpha 0\rangle + \beta 1\rangle$

Final Remarks on Quantum Teleportation!

- Teleportation is not an application of quantum information — it's a way to perform *quantum communication*.
- Teleportation motivates *entanglement distillation* as a means to reliable quantum communication.

Applications: Building the Quantum Internet

Quantum Teleportation is the "Protocol" for future global networks:

- **Solving the No-Cloning Problem:** Since qubits cannot be copied, teleportation is the *only* way to move a quantum state from Alice to Bob.
- **Quantum Repeaters:** Over long distances, fiber optic signals fade. Teleportation allows us to "hop" the state from one station to the next without losing data.
- **Absolute Security:** The "message" never physically travels. A hacker intercepting the classical bits sees only random noise.
- **Distributed Computing:** Connecting small quantum computers in different cities to act as one giant super-processor.

The "Atomic Unit"

Just as TCP/IP is the foundation of the modern internet, **Teleportation + Entanglement** is the foundation of the Quantum Internet.

Superdense Coding: Concept & Importance

What is Superdense Coding?

- A quantum communication protocol that allows **sending 2 classical bits using only 1 qubit**.
- Exploits **entanglement** between two parties (commonly called Alice and Bob).
- The sender (Alice) applies a quantum operation on her qubit and sends it to the receiver (Bob), who measures both qubits to retrieve the classical information.

Why is it important?

- Demonstrates the **power of entanglement** in enhancing communication efficiency beyond classical limits.
- Forms a foundational concept for **quantum communication networks** and quantum information theory.
- Highlights potential for **secure and high-capacity communication** in future quantum technologies.

Superdense Coding Set-Up

Scenario

Alice has *two classical bits* that she wishes to transmit to Bob.

- Alice is able to send a *single qubit* to Bob.
- Alice and Bob *share an e-bit*.

Remark

Without the e-bit, Alice and Bob's task would be impossible...

Holevo's theorem implies that two classical bits of communication cannot be reliably transmitted by a single qubit alone.

Superdense Coding: 2 Bits, 1 Qubit

The Goal: Send **two classical bits** by sending only **one physical qubit**.

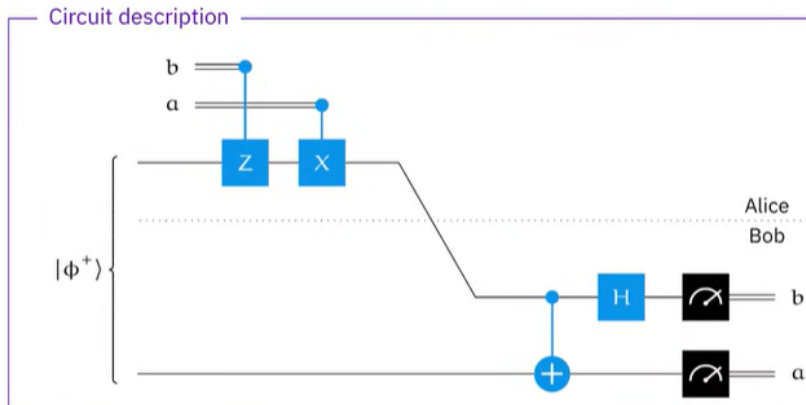
- **Holevo's Theorem:** Normally, 1 qubit = 1 bit of info.
- **The Loophole:** Because Alice and Bob share an **e-bit**, they already have "half the bridge" built.
- **Efficiency:** This protocol **doubles the bandwidth** of the communication line.

What if we want to send 4 bits?

- **Resources Needed:** 2 shared e-bits.
- **Physical Action:** Alice sends only **2 qubits**.
- **Final Result:** Bob extracts **4 classical bits**.

Quantum Entanglement acts as a hidden multiplier for communication!

Superdense Coding Circuit Diagram



Superdense Coding: The Setup

Step 1: The Initial State

Alice and Bob share a Bell State: $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$.

- **Left Qubit (Top Line):** Alice's half.
- **Right Qubit (Bottom Line):** Bob's half.

Superdense Coding: Alice's Encoding

Step 2: Alice's Encoding Table

Message	Alice's Gate	New Shared State
00	I (None)	$ \Phi^+\rangle$
01	X (Bit-flip)	$ \Psi^+\rangle$
10	Z (Phase-flip)	$ \Phi^-\rangle$
11	ZX (Both)	$ \Psi^-\rangle$

Step 3: Transmission & Readout

Alice sends her **1 qubit** to Bob. Bob now has **both**. He performs a **Bell Measurement** to decode the 2 bits.

Superdense coding analysis

$$|\phi^+\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$

$$|\phi^-\rangle = \frac{1}{\sqrt{2}}|00\rangle - \frac{1}{\sqrt{2}}|11\rangle$$

$$|\psi^+\rangle = \frac{1}{\sqrt{2}}|01\rangle + \frac{1}{\sqrt{2}}|10\rangle$$

$$|\psi^-\rangle = \frac{1}{\sqrt{2}}|01\rangle - \frac{1}{\sqrt{2}}|10\rangle$$

a b	Alice's action	Bob's action
00	$ \phi^+\rangle \mapsto \phi^+\rangle$	$ \phi^+\rangle \mapsto 00\rangle$
01	$ \phi^+\rangle \mapsto \phi^-\rangle$	$ \phi^-\rangle \mapsto 01\rangle$
10	$ \phi^+\rangle \mapsto \psi^+\rangle$	$ \psi^+\rangle \mapsto 10\rangle$
11	$ \phi^+\rangle \mapsto \psi^-\rangle$	$ \psi^-\rangle \mapsto - 11\rangle$

Final Remarks on Superdense coding Protocol!

- Superdense coding seems unlikely to be useful in a practical sense.
- The underlying idea is fundamentally important, and illustrates an interesting aspect of entanglement.

Lecture Summary: Key Concepts Covered I

- **Quantum Circuits:** Representation of quantum evolution; wires = qubits, gates = unitary operations. Local gates act on single qubits, global gates (e.g., CNOT) enable **entanglement**.
- **Inner Products & Geometry:** Overlap between states $\langle \phi | \psi \rangle$. Orthogonal states: $\langle \phi | \psi \rangle = 0$, orthonormal basis: $\langle v_i | v_j \rangle = \delta_{ij}$.
- **Projections:** Project a state onto a specific subspace: $P = |\psi\rangle\langle\psi|$.
- **Compound Systems & Tensor Products:** Multi-qubit states in $(\mathbb{C}^2)^{\otimes n}$; lexicographical ordering: $|00\rangle, |01\rangle, |10\rangle, |11\rangle$; joint operations use Kronecker product $M \otimes N$.
- **Bilinearity:** Tensor products linear in every slot:
 $\alpha(|A\rangle \otimes |B\rangle) = (\alpha|A\rangle) \otimes |B\rangle$.
- **Probabilistic Operations:** Compound system transformations via stochastic matrices.

Lecture Summary: Key Concepts Covered II

- **Quantum Teleportation:** Transfer of an unknown qubit state using entanglement and classical communication. Fundamental for secure quantum communication.
- **Superdense Coding:** Send 2 classical bits using 1 entangled qubit — demonstrates quantum advantage in communication.

Next Lecture: Deep Dive into CHSH Game Strategy and Analysis